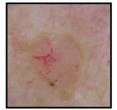
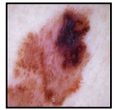
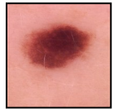
ISIC 2019 Dataset


BCC
3,323


Melanoma
4,522

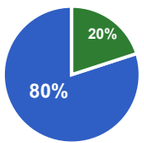

Nevus
4,600

12,445 images, 3 classes

Preprocessing

Resize 224 × 224 → Augment flip, rotate, zoom → Normalize 0–1

Data Split



Training 80%
9,956 images

Validation 20%
2,489 images

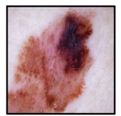
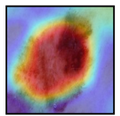
Stratified, image-level

Branch 1: MobileNetV2 (CNN, ~2.6M parameters)

Architecture

Input 224 × 224 → MobileNetV2 (ImageNet) → Global Avg Pooling → Dense 256 ReLU → Dense 64 ReLU → Dense 3 Softmax

Grad-CAM explainability (melanoma)


→


Baseline FP32 evaluation

MobileNetV2 FP32

Accuracy: 90.96%
Macro F1: 91.27%
Size: 30.21 MiB
CPU latency: 104.89 ms

Optimization experiments

Dynamic INT8 TFLite: 90.88%

Full INT8 TFLite: 90.72%
Recommended edge candidate

Pruning (50%): 83.49%

Accuracy on the validation split

Independent branches: no ensemble or prediction fusion

Branch 2: DINOv2 (Vision Transformer, ~86.6M parameters)

Architecture

Input 224 × 224 → Patch embed 256 + CLS → Transformer encoder 12 blocks → CLS token → Linear head 768 → 3 → Softmax

Baseline FP32 evaluation

DINOv2 FP32

Accuracy: 86.66%
Macro F1: 86.82%
Size: 330.34 MiB
CPU latency: 373.35 ms

Optimization experiments

Dynamic INT8: 61.59%

Folded INT8: 61.59%

INT8 not recommended
(significant accuracy drop)

Classification Output


BCC (Class 0)


Melanoma (Class 1)


Nevus (Class 2)


Raspberry Pi 4
INT8: 13.9 ms